ZOMATO | SQL + POWER BI PROJECT

Restaurant Ratings & Order Trends Analysis

Turning order, customer, restaurant and review data into interactive dashboards using MySQL and Power BI

Presented by Gayathri
Anudip Foundation • Course: DBA_AI

Project Overview

Industry

Food delivery and restaurant marketplace analytics

Real-world problem

Order, customer, restaurant and review data are hard to assess together

Purpose

Compare operational and customer-facing indicators in one place

Approach

MySQL for storage and SQL analysis; Power BI for modeling, DAX and dashboards

Tools & Technologies

MySQL
Database: zomato_analytics

SQL
Cleaning and analysis

Power BI Desktop
Report and dashboards

Power Query
Import and transformation

DAX
Measures and date table

CSV files
Four source datasets

Problem Statement & Objectives

Problem Statement

Bring order, customer, restaurant and review information into a single analytical workflow. By preparing data in MySQL and visualizing it in Power BI, the report helps explore order trends, delivery indicators, customer segments and restaurant ratings.

Analysis is descriptive: it shows patterns, not cause and effect.

Project Objectives

1. Prepare and analyze multi-table data using SQL
2. Build a relational data model in Power BI
3. Create DAX measures for orders, value, delivery and customers
4. Design interactive dashboards for orders, customers, ratings and locations
5. Summarize insights and suggest follow-up actions

Business Questions We Set Out to Answer

Volume & value

How many orders are recorded and what is their total order value?

Order status

How are orders split between Delivered, Cancelled and Returned?

City, time & segment

How do order volumes vary by city, hour and customer segment?

Delivery & basket size

What are the average order value and average delivery time?

Customers

How are customers spread across New, Regular and Frequent segments?

Ratings & cuisine

Which restaurants rate highest, and how do cuisines and reviews compare?

Dataset Overview

customers.csv

900

rows • 4 columns

customer_id, customer_city, signup_date, customer_segment

orders.csv

12,000

rows • 10 columns

order_id, customer_id, restaurant_id, order_date, order_time, order_amount, delivery_time_mins, order_status …

restaurants.csv

250

rows • 11 columns

restaurant_id, restaurant_name, city, primary_cuisine, restaurant_rating, online_delivery …

reviews.csv

4,300

rows • 6 columns

review_id, restaurant_id, customer_id, review_date, rating, review_text

17,450 rows • 31 columns across four CSV files loaded into the MySQL database zomato_analytics (practice dataset)

Project Workflow

1

Industry Selection

Food delivery marketplace

2

Problem Identification

Scattered order and review data

3

Dataset Collection

Four CSV files

4

MySQL Database Setup

zomato_analytics

5

Cleaning & Transformation

Using SQL

6

SQL Data Analysis

Aggregations and joins

7

Power BI Data Modeling

Relationships and DAX

8

Dashboard Development

Five report pages

9

Insights & Recommendations

Findings and next steps

Result

5-page interactive Power BI report

MySQL Database & SQL Analysis

- **Setup**
  CREATE DATABASE / CREATE TABLE for four tables
- **Import**
  Load CSV rows into MySQL
- **Inspect & filter**
  SELECT, WHERE, ORDER BY, LIMIT
- **KPIs**
  COUNT, SUM, AVG aggregates
- **Summaries**
  GROUP BY status, city, cuisine, segment
- **Combine tables**
  JOIN on customer_id and restaurant_id

Order status summary

```
SELECT order_status,
       COUNT(*) AS order_count,
       SUM(order_amount) AS total_order_value
FROM orders
GROUP BY order_status
ORDER BY order_count DESC;
```

Top 10 restaurants by order value (JOIN)

```
SELECT r.restaurant_name,
       SUM(o.order_amount) AS total_value,
       COUNT(DISTINCT o.order_id) AS total_orders
FROM orders AS o
JOIN restaurants AS r
  ON o.restaurant_id = r.restaurant_id
GROUP BY r.restaurant_name
ORDER BY total_value DESC
LIMIT 10;
```

Power BI Data Preparation & Model

Import

Load from MySQL or the prepared CSV files

Set data types

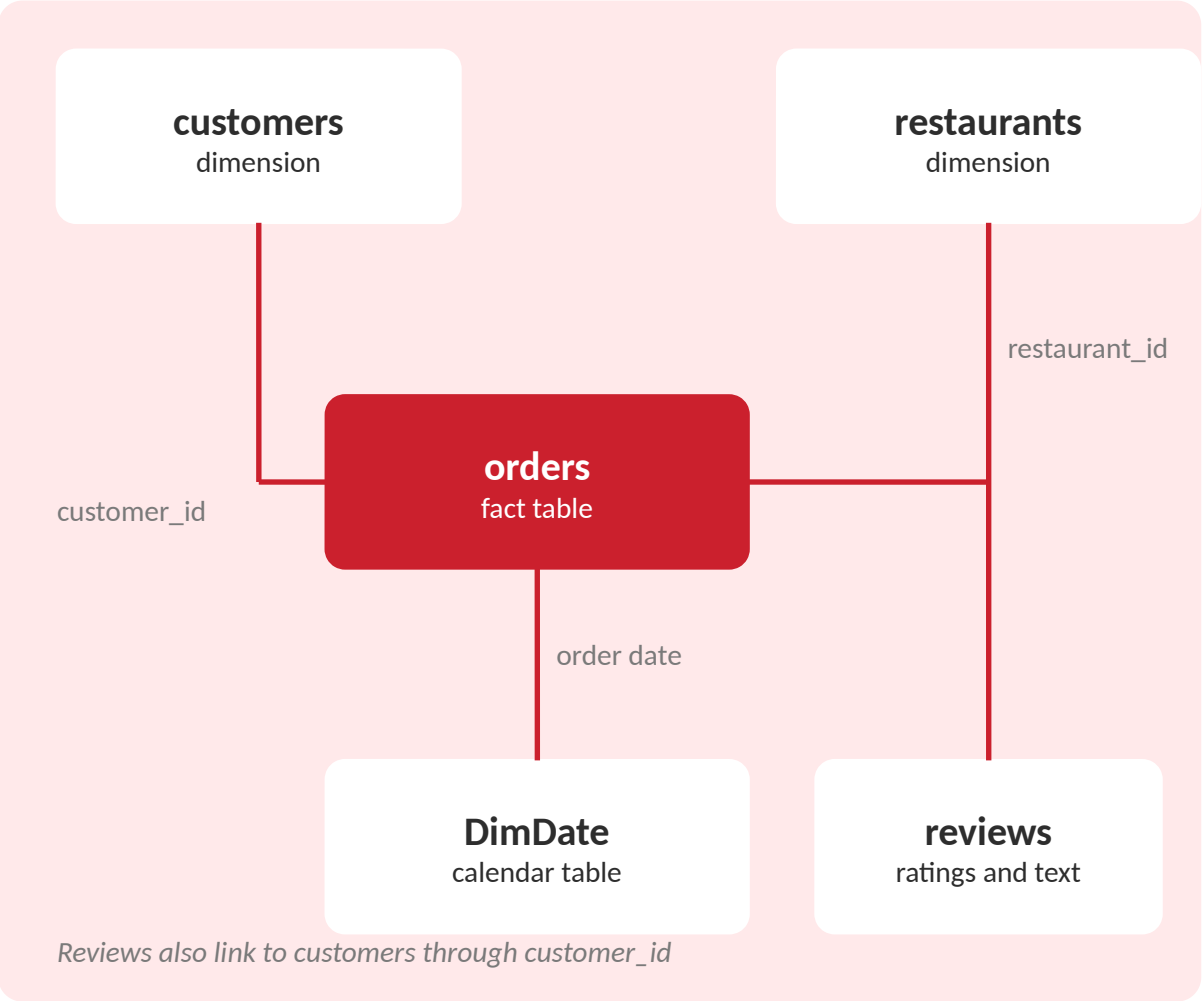
IDs as identifiers, amounts as decimals, dates and times as Date/Time

Clean in Power Query

Check blanks and inconsistent labels; keep raw data unchanged

Model

One-to-many relationships plus a date table



DAX Measures

Total Orders

```
DISTINCTCOUNT(
  orders[order_id])
```

Unique orders recorded

Total Order Value

```
SUM(
  orders[order_amount])
```

Sum of order amounts

Cancellation Rate %

```
DIVIDE(
  [Cancelled Orders],
  [Total Orders], 0)
```

Cancelled share of all orders

Average Order Value

```
DIVIDE(
  [Total Order Value],
  [Total Orders], 0)
```

Value per order

Average Delivery Time

```
AVERAGE(
  orders[delivery_time_mins])
```

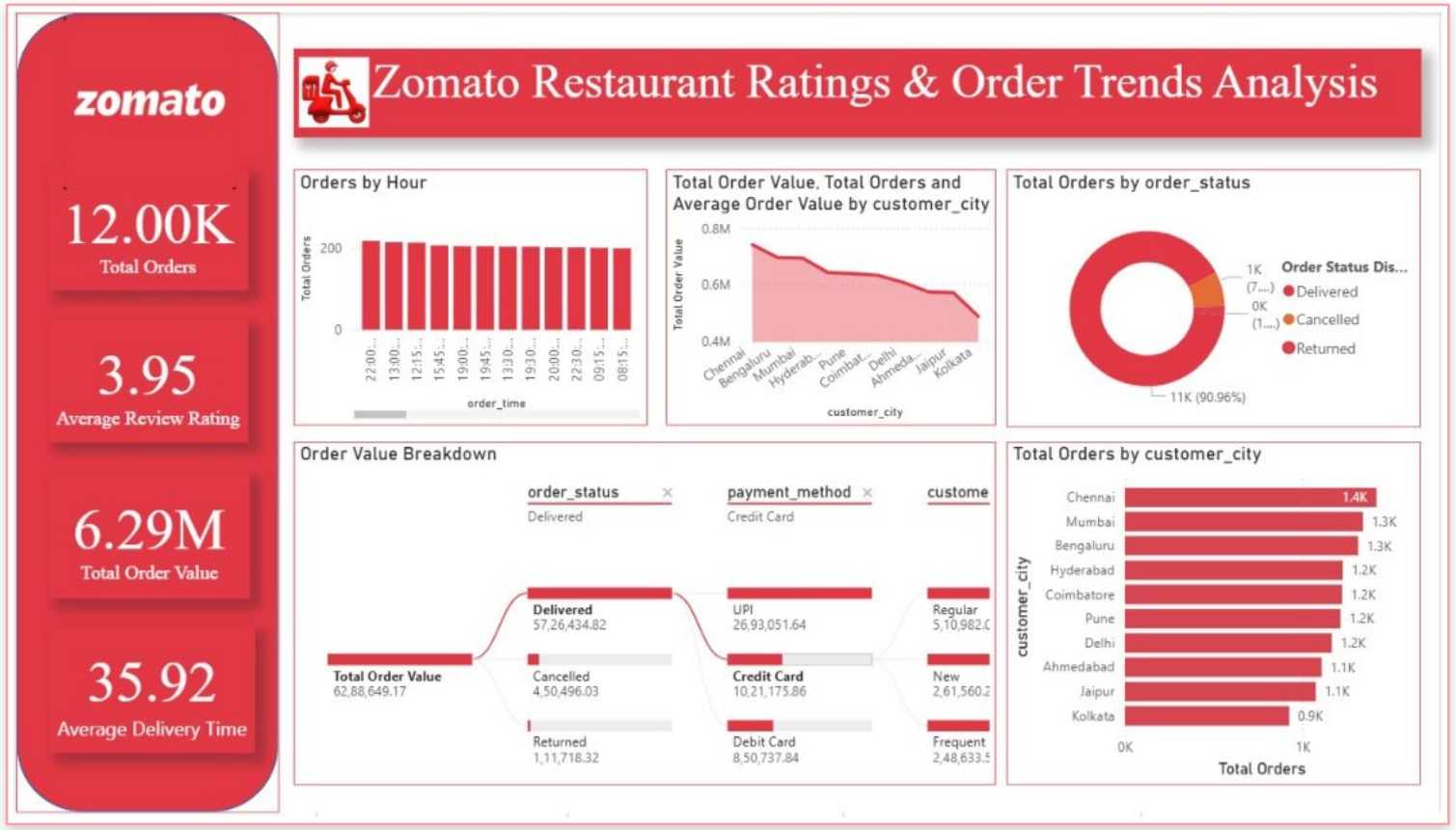
Mean delivery minutes

Segment Customers

```
CALCULATE(
  [Total Customers],
  customers[customer_segment]
    = "Regular")
```

New, Regular, Frequent counts

Dashboard 1: Ratings & Order Trends Overview



Page 1 of 5: KPI cards, order timing, status mix and city ranking

12.00K
Total Orders

3.95
Average Review Rating

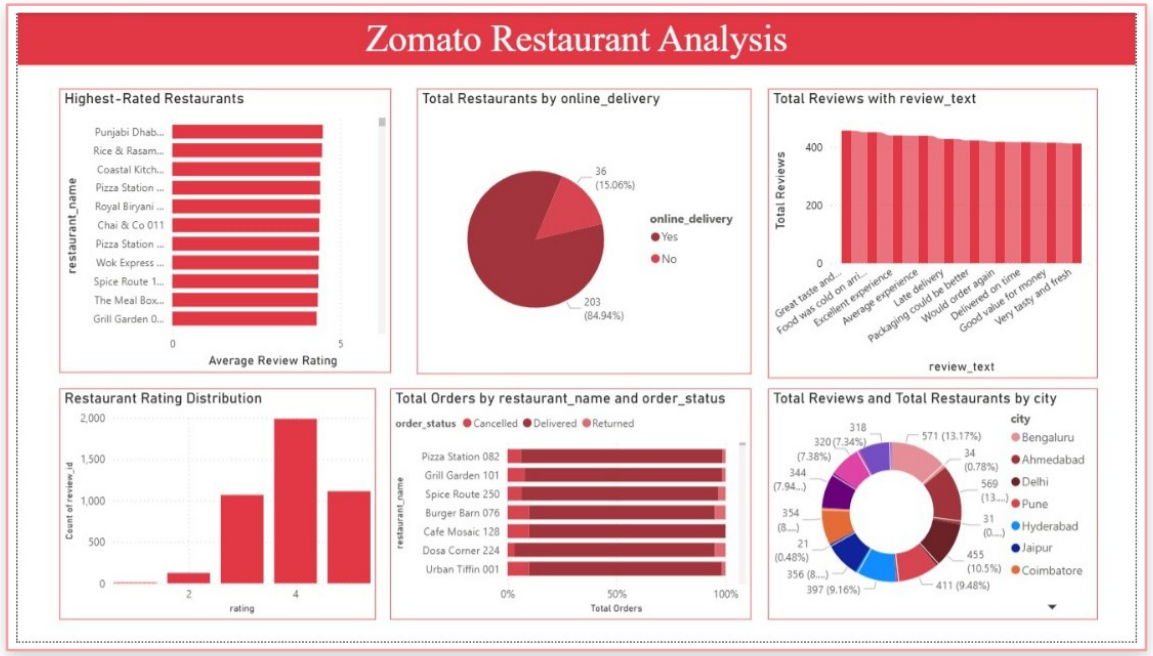
6.29M
Total Order Value

35.92
Avg Delivery Time

- Orders by hour compared across time slots
- Status donut: Delivered is the largest share
- Decomposition tree: value by status, payment, segment
- Cities ranked by total orders

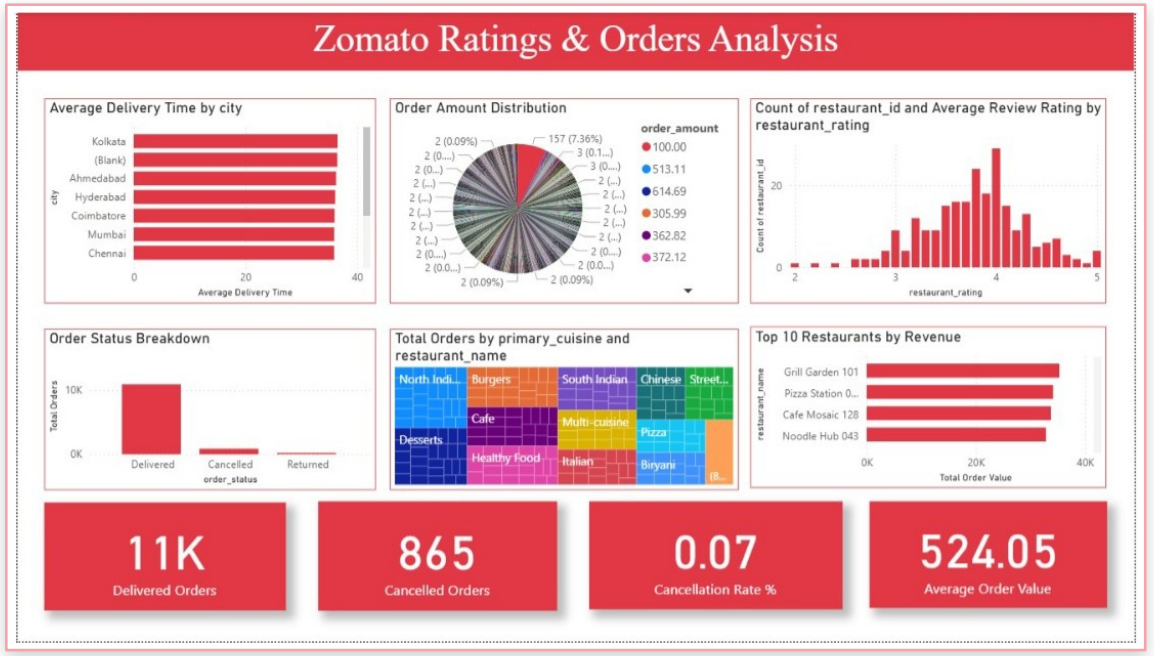
Dashboards 2 & 3: Restaurants, Ratings & Orders

Restaurant Analysis



- Highest-rated restaurants and rating distribution
- Online delivery: 203 Yes vs 36 No
- Reviews and restaurants by city

Ratings & Orders Analysis



- Delivery time by city, top 10 restaurants by value
- Cuisine treemap and status breakdown
- Cards: 865 cancelled orders, average order value 524.05

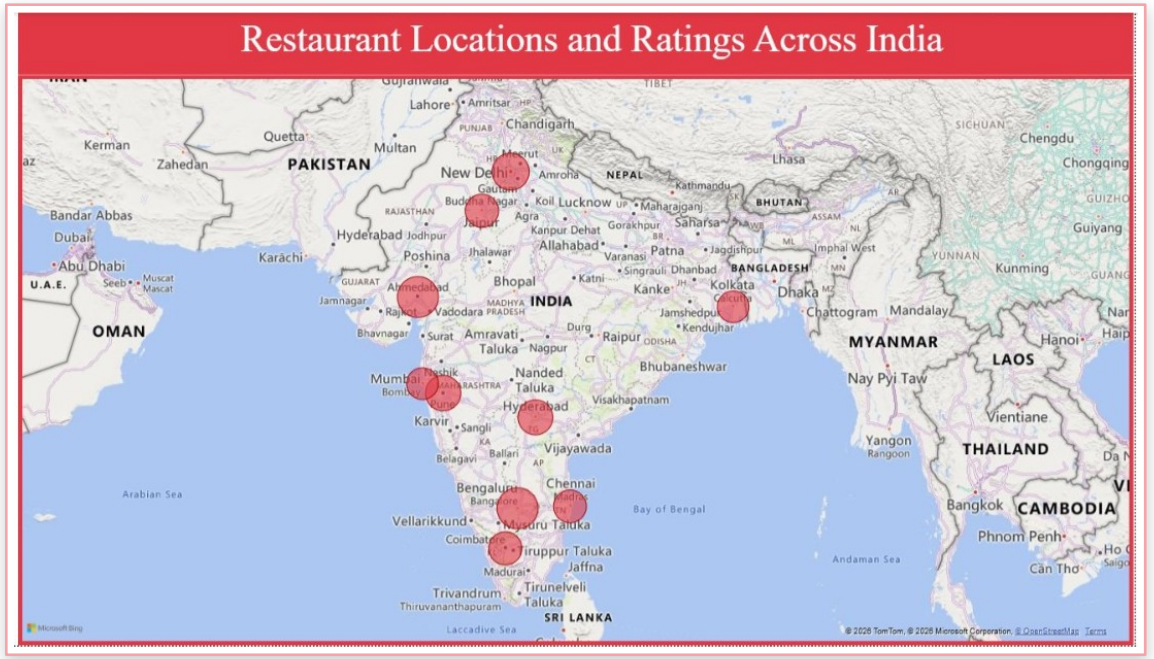
Dashboards 4 & 5: Customers and Locations

Customer Behaviour & Insights



- 900 customers: 439 Regular, 232 New, 229 Frequent
- Customer distribution and segments by city
- Average spend by segment

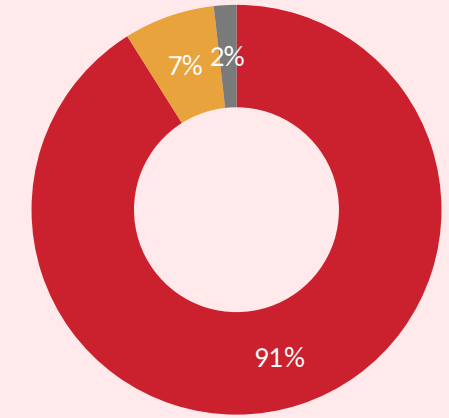
Restaurant Locations Across India



- Bubble map of restaurant locations
- Tooltips: city, reviews, rating, total orders
- Geographic view of ratings and activity

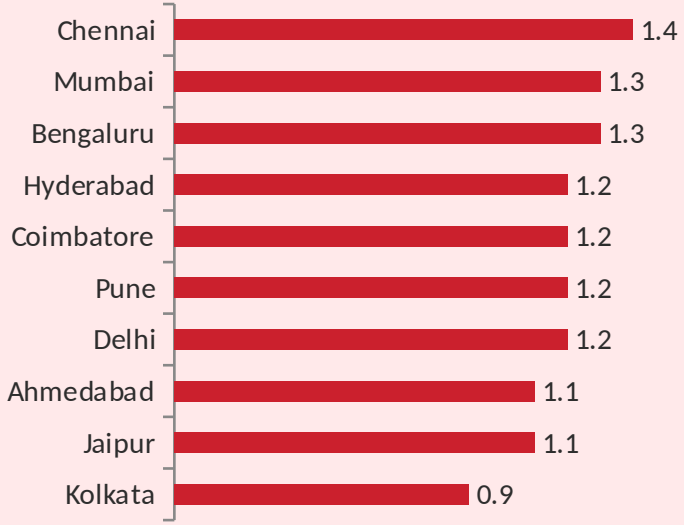
Key Insights from the Dashboards

Order value by status



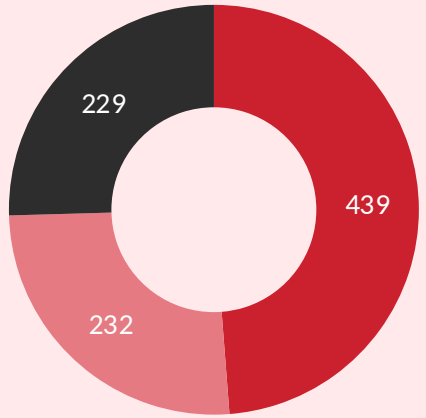
Delivered orders carry about 91% of the 6.29M total order value

Total orders by city (K)



Chennai leads on orders; Kolkata is lowest of the ten cities

Customers by segment



Regular customers are the largest of the 900 (439)

Ratings cluster near 4 stars (average 3.95); 203 restaurants offer online delivery vs 36 that do not; 865 orders were cancelled (0.07 rate)

Challenges Faced & How We Overcame Them

Challenge	How it was addressed
(Blank) Year Month in the trend chart	Built a DimDate table, created a date-only key, kept the relationship active and sorted by YearMonthSort
Hours and months sorted alphabetically	Used a sort-by-column field so time labels follow chronological order
KPI definitions: Repeat vs Frequent, retention 0.25, cancellation 0.07	Kept one segment definition, used DIVIDE with the correct denominator and applied percentage formatting
Crowded visuals: order amounts and review text	Replaced with bins or histograms and used Top N filters with shorter labels
Source, currency and metric validation	Matched Power BI totals with SQL results and labeled the data as practice data with units unconfirmed

Conclusion

Key Takeaways

- MySQL and Power BI combined into one analytical workflow
- Five dashboard pages compare orders, customers, ratings and locations
- Delivered orders dominate; cancellations and returns are small but worth tracking
- Findings are descriptive and need validated sources and units

Deliverables on GitHub: PBIX report • SQL script • four CSV datasets • dashboard screenshots • README

Thank you